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In [1]: # Program 5: Detect outliers with the IQR method
import pandas as pd

FILE = r"D:\Datascience-MachineLearning\Datasets\titanic.csv" # <-- change to your file path
COLUMN = "Fare" # numerical column to check

df = pd.read_csv(FILE)

q1 = df[COLUMN].quantile(0.25)
q3 = df[COLUMN].quantile(0.75)
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr

print("Q1 =", q1, " Q3 =", q3, " IQR =", iqr)
print("Lower limit =", lower, " Upper limit =", upper)

outliers = df[(df[COLUMN] < lower) | (df[COLUMN] > upper)]
print("\nNumber of outliers:", len(outliers))
print(outliers)

clean = df[(df[COLUMN] >= lower) & (df[COLUMN] <= upper)]
print("\nData after removing outliers:", clean.shape)
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Q1 = 7.9104 Q3 = 31.0 IQR = 23.0896
 Lower limit = -26.724 Upper limit = 65.6344

Number of outliers: 116

	PassengerId	Survived	Pclass	\
1	2	1	1	
27	28	0	1	
31	32	1	1	
34	35	0	1	
52	53	1	1	
..	...	...	...	
846	847	0	3	
849	850	1	1	
856	857	1	1	
863	864	0	3	
879	880	1	1	

	Name	Sex	Age	SibSp	\
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
27	Fortune, Mr. Charles Alexander	male	19.0	3	
31	Spencer, Mrs. William Augustus (Marie Eugenie)	female	NaN	1	
34	Meyer, Mr. Edgar Joseph	male	28.0	1	
52	Harper, Mrs. Henry Sleeper (Myna Haxtun)	female	49.0	1	
..	...	...	...	...	
846	Sage, Mr. Douglas Bullen	male	NaN	8	
849	Goldenberg, Mrs. Samuel L (Edwiga Grabowska)	female	NaN	1	
856	Wick, Mrs. George Dennick (Mary Hitchcock)	female	45.0	1	
863	Sage, Miss. Dorothy Edith "Dolly"	female	NaN	8	
879	Potter, Mrs. Thomas Jr (Lily Alexenia Wilson)	female	56.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
1	0	PC 17599	71.2833	C85	C
27	2	19950	263.0000	C23 C25 C27	S
31	0	PC 17569	146.5208	B78	C
34	0	PC 17604	82.1708	NaN	C
52	0	PC 17572	76.7292	D33	C
..	...	...	...	...	...
846	2	CA. 2343	69.5500	NaN	S
849	0	17453	89.1042	C92	C
856	1	36928	164.8667	NaN	S
863	2	CA. 2343	69.5500	NaN	S

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879      1      11767  83.1583      C50      C
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[116 rows x 12 columns]
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Data after removing outliers: (775, 12)
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In [ ]:
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